variables, perhaps representing the positions of particles. These extra quantities are conventionally called *hidden variables*, although some proponents of this approach don't like the label, as it's these variables that we actually observe when we make a measurement. We can just call them particles, since that's the case that is usually considered. The wave function then takes on the role of a *pilot wave*, guiding the particles as they move around. It's like particles are little floating barrels, and the wave function describes waves and currents in the water that push the barrels around. The wave function obeys the ordinary Schrödinger equation, while a new "guidance equation" governs how it influences the particles. The particles are guided to where the wave function is large, and away from where it is nearly zero.

The first such theory was presented by Louis de Broglie, at the 1927 Solvay Conference. Both Einstein and Schrödinger were thinking along similar lines at the time. But de Broglie's ideas were harshly criticized at Solvay, by Wolfgang Pauli in particular. From the records of the conference, it seems as if Pauli's criticisms were misplaced, and de Broglie actually answered them correctly. But he was sufficiently discouraged by the reception that de Broglie abandoned the idea.

In a famous book from 1932, *Mathematical Foundations of Quantum Mechanics*, John von Neumann proved a theorem about the difficulty of constructing hidden-variable theories. Von Neumann was one of the most brilliant mathematicians and physicists of the twentieth century, and his name carried enormous credibility among researchers in quantum mechanics. It became standard practice, whenever anyone would suggest that there might be a more definite way to formulate quantum theory than the vagueness inherent in the Copenhagen approach, for someone to invoke the name of von Neumann and the existence of his proof. That would squelch any budding discussion.

In fact what von Neumann had proven was something a bit less than most people assumed (often without reading his book, which wasn't translated into English until 1955). A good mathematical theorem establishes a result that follows from clearly stated assumptions. When we would like to invoke such a theorem to teach us something about the real world, however, we have to be very careful that the assumptions are actually true in reality. Von Neumann made assumptions that, in retrospect, we don't have to make if our task is to invent a theory that reproduces the